HapticSense — Adaptive Haptic Navigation Belt for Visually Impaired Users

Sindhu Priya Nagendra Rao
Steve Sanghi College of Engineering
Northern Arizona University
Flagstaff, Arizona
Sn2388@nau.edu

***Abstract*—This paper reviews wearable obstacle detection systems and vibrotactile belt interfaces for visually impaired navigation and formulates research questions and hypotheses for evaluating a low-cost adaptive haptic navigation belt.**

***Index Terms*—Wearable assistive technology, haptic feedback, vibrotactile belt, obstacle detection, ultrasonic sensor, visually impaired navigation**

I. Literature Review

A. Wearable Obstacle Detection Systems

Wearable assistive technologies for visually impaired navigation have increasingly relied on embedded sensing platforms to provide real-time environmental awareness. Ultrasonic-based obstacle detection remains one of the most practical approaches due to its low cost, low power consumption, and computational simplicity. Olomowewe *et al.* developed a wearable obstacle detection and warning device using ultrasonic sensing and vibration-based proximity alerts, demonstrating reliable obstacle detection within predefined distance thresholds [3]. Similarly, Sharom *et al.* implemented an Arduino-based multi-sensor wearable system capable of 360-degree obstacle detection using multiple ultrasonic sensors and directional vibration motors [1]. These works validate the feasibility of low-cost microcontroller-based assistive systems and highlight the practicality of distributed sensing architectures. However, both systems primarily emphasize detection capability and hardware integration, with limited quantitative analysis of haptic encoding efficiency, recognition latency, or perceptual optimization.

B. Vibrotactile Belt Interfaces and Encoding Strategies

In contrast to sensing-focused implementations, Cosgun *et al.* investigated the human factors associated with vibrotactile belt interfaces for navigation assistance [2]. Their eight-motor tactile belt design evaluated both directional and rotational vibration patterns through controlled user studies, measuring recognition accuracy and reaction time. Results indicated that discrete two-pulse directional patterns yielded faster response times and higher user preference compared to continuous vibrations. Rotational cues demonstrated near-perfect recognition accuracy, suggesting that waist-based spatial mapping is effective for orientation control. Flores *et al.* further demonstrated the effectiveness of vibrotactile guidance systems for wayfinding of blind walkers, reinforcing the reliability of spatially distributed vibration feedback [4]. Although these studies validate perceptual encoding effectiveness, they do not integrate real-time environmental sensing into the belt architecture.

C. State of the Art and Comparative Analysis

The current state of the art therefore consists of two complementary but largely independent research directions. The first focuses on sensor-driven wearable systems capable of detecting obstacles and generating proximity-based alerts in real time [1], [3]. These systems demonstrate technical feasibility and real-world deployment potential but often employ relatively simple vibration mappings without rigorous perceptual validation. The second direction emphasizes haptic interface optimization and encoding strategy evaluation using tactile belts [2], [4], providing strong experimental evidence for effective directional and rotational cue design. While both approaches advance assistive navigation technology, they remain insufficiently integrated.

D. Identified Research Gap

Despite significant progress in both embedded sensing and vibrotactile interface research, limited work combines adaptive obstacle detection with empirically validated directional encoding strategies in a unified wearable belt system. Existing sensing platforms provide obstacle presence awareness but do not optimize vibration patterns for rapid spatial interpretation [1], [3]. Conversely, tactile belt studies evaluate encoding performance but omit real-time sensing integration and adaptive feedback modulation [2], [4]. Additionally, few studies explore adaptive intensity mapping, where vibration strength or frequency dynamically scales with obstacle proximity to enhance intuitive interpretation [3]. This separation highlights a clear integration gap within the current literature.

E. Motivation for the Proposed System

The proposed HapticSense system is motivated by the need to bridge this integration gap by combining low-cost ultrasonic sensing with validated vibrotactile encoding strategies in a belt-based wearable form factor. By leveraging empirical findings on directional and rotational cue effectiveness [2], [4] while incorporating real-time obstacle detection architectures demonstrated in prior systems [1], [3], the project aims to

improve recognition accuracy, reduce reaction latency, and enhance spatial awareness without increasing auditory cognitive load. This integrated approach advances the current state of the art toward a practical, adaptive, and user-centered wearable navigation system.

F. Design Considerations for Integrated Haptic Navigation

The reviewed studies highlight two essential dimensions in wearable navigation systems: reliable obstacle detection and perceptually effective haptic encoding. Ultrasonic-based platforms demonstrate feasibility in real-time distance measurement and obstacle awareness [1], [3], while tactile belt research establishes the importance of spatially aligned vibration patterns for rapid directional interpretation [2], [4].

II. Research Questions and Hypotheses

A. Primary Research Question

Can a low-cost, two-sensor haptic belt provide reliable directional obstacle awareness for visually impaired users using proximity-based vibration feedback, building upon prior ultrasonic wearable systems [1], [3] and validated vibrotactile encoding strategies [2], [4]?

Primary Hypothesis (H1)

A waist-mounted haptic belt using two ultrasonic sensors and directional vibration motors will enable users to correctly identify obstacle direction (left vs. right) with accuracy significantly greater than chance (50%), consistent with spatial encoding effectiveness demonstrated in vibrotactile belt studies [2], [4].

Expected Result

Users will correctly identify obstacle direction in at least 80% of trials, reflecting effective waist-based spatial mapping as observed in prior tactile belt research [2].

Alternative Result

Users may struggle to distinguish directional cues, resulting in performance close to chance level (50%), suggesting insufficient perceptual discrimination despite prior encoding findings [2].

Interpretation

This hypothesis extends validated vibrotactile spatial encoding principles [2], [4] into a simplified real-time sensing architecture inspired by ultrasonic wearable systems [1], [3].

Primary Hypothesis (H2)

Proximity-based modulation of vibration intensity (stronger vibration for closer obstacles) will reduce reaction time compared to fixed-intensity vibration feedback, building upon proximity-alert mechanisms demonstrated in ultrasonic wearable systems [3].

Expected Result

Reaction times will decrease as vibration intensity scales with decreasing obstacle distance, consistent with urgency signaling mechanisms discussed in wearable obstacle detection research [3].

Alternative Result

Reaction time may not significantly differ between adaptive and fixed-intensity modes, suggesting that intensity scaling does not meaningfully improve interpretability beyond baseline proximity alerts [1].

Primary Hypothesis (H3)

Simultaneous activation of both vibration motors (indicating frontal or dual-side obstacle presence) will be interpreted as a "stop or slow" cue more consistently than single-side activation patterns, consistent with multi-motor spatial encoding principles demonstrated in tactile belt research [2], [4].

Expected Result

Participants will demonstrate consistent behavioral hesitation or stopping when both motors activate, reflecting effective multi-point spatial cue integration [2].

Alternative Result

Participants may interpret dual activation inconsistently, suggesting ambiguity in multi-motor cue design despite prior spatial encoding validation [2].

B. How the Device Enables Investigation

The proposed HapticSense belt enables investigation of these hypotheses by:

- Collecting ultrasonic distance measurements from left and right sensors in real time [1]
- Mapping sensor data directly to spatially aligned vibration motors following validated encoding strategies [2]
- Logging user response time and directional accuracy during controlled obstacle trials
- Comparing performance under fixed-intensity vs. adaptive-intensity vibration conditions inspired by prior wearable obstacle detection systems [3]

Because the system isolates only two sensors and two motors, it allows controlled evaluation of directional mapping and adaptive feedback without confounding complexity from multi-sensor arrays.

III. Design and Implementation

The proposed HapticSense system integrates real-time obstacle detection with directional vibrotactile feedback using a compact and low-cost embedded architecture. The system is designed to provide intuitive spatial awareness through a waist-mounted configuration consisting of ultrasonic sensors and vibration motors.

A. System Architecture

The hardware architecture consists of:

- Two ultrasonic sensors (HC-SR04) positioned on the left and right sides
- Two vibration motors corresponding to directional feedback
- An Arduino-compatible microcontroller (SparkFun RedBoard)
- NPN transistors for motor driving
- Base resistors for current limiting
- Flyback diodes for motor protection

Each ultrasonic sensor continuously measures distance to nearby obstacles and transmits the data to the microcontroller. Based on the measured proximity, the system activates the corresponding vibration motor to indicate obstacle direction.

B. Circuit Design and Component Integration

The circuit design follows a modular approach where sensing and actuation components are interfaced through the microcontroller.

Each vibration motor is controlled using an NPN transistor acting as a low-side switch:

- Collector → Connected to motor negative terminal
- Emitter → Connected to ground
- Base → Connected to Arduino digital pin through a resistor (220Ω–1kΩ)

Since motors are inductive loads, flyback diodes are placed across each motor to protect the circuit from voltage spikes:

- Cathode (striped side) → Connected to +5V
- Anode → Connected to motor negative terminal (transistor side)

This configuration ensures safe operation and prevents damage to the transistor and microcontroller.

C. Sensor-to-Actuator Mapping

The system uses a direct mapping strategy between sensors and motors:

- Left sensor → Left vibration motor
- Right sensor → Right vibration motor

When an obstacle is detected within a predefined threshold distance, the corresponding motor is activated. Vibration intensity can also be modulated based on proximity to provide adaptive feedback. Simultaneous activation of both motors represents frontal obstacle presence, aligning with spatial encoding principles discussed in prior work.

D. Implementation Workflow

The implementation follows these steps:

1) Ultrasonic sensors measure distance using trigger and echo signals
2) Arduino processes distance data in real time
3) Threshold logic determines obstacle presence
4) Corresponding transistor is switched ON
5) Vibration motor is activated to provide haptic feedback

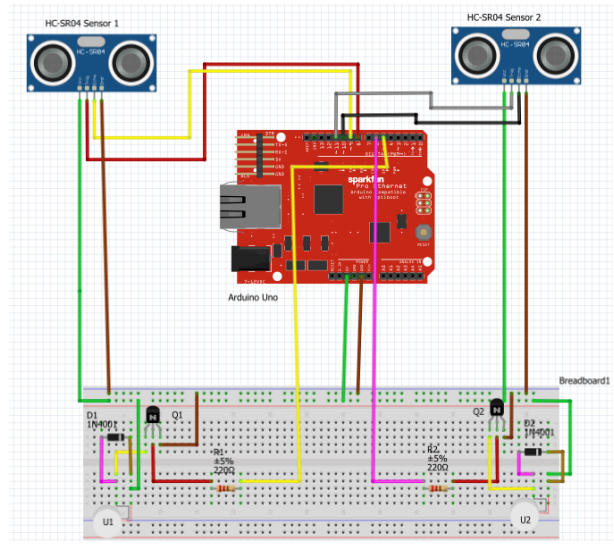
This workflow ensures low latency and real-time responsiveness, which is critical for assistive navigation systems.

E. Prototype Circuit Layout

The complete wiring layout of the system, including sensors, transistors, resistors, motors, and protection diodes, is shown in Fig. 1.

F. Design Justification

The design is based on low-cost, widely available components and focuses on simplicity and scalability. By combining ultrasonic sensing with directional haptic feedback, the system bridges the gap between obstacle detection and perceptually effective feedback, as identified in the literature review.



[t]

Fig. 1. Circuit design of the HapticSense system showing ultrasonic sensors, transistor-based motor drivers, and flyback diode protection.

IV. Methods Plan

This section describes the experimental methodology that will be used to evaluate the proposed HapticSense system. The methods are designed to systematically assess the ability of the device to detect obstacle direction, estimate obstacle distance, and deliver spatially meaningful vibrotactile feedback under controlled conditions. The study focuses on repeatable measurement, controlled testing, and quantitative evaluation of the hypotheses defined in Section II.

A. Experimental Design Framework

This study employs a controlled within-subject experimental design to evaluate the performance of a two-sensor vibrotactile navigation system. Each participant is exposed to multiple trials under varying obstacle and feedback conditions in order to reduce inter-subject variability and improve sensitivity to system-level effects.

The independent variables are: 1) obstacle direction (left, right, and dual/front), 2) obstacle distance (e.g., 20 cm, 40 cm, 60 cm, and 80 cm), and 3) feedback mode (fixed-intensity vibration and adaptive-intensity vibration). The dependent variables are: 1) directional accuracy, 2) reaction time, and 3) perceptual consistency of feedback interpretation. Control variables include sensor placement, indoor testing environment, participant posture, and obstacle presentation procedure.

B. Participants

The initial prototype evaluation will involve a small sample of participants ($n = 1$–5) for feasibility testing and proof-of-concept validation. Participants will be healthy adults without known tactile sensory impairments. Since this stage of the project is primarily focused on validating system functionality and experimental feasibility, a limited participant pool is appropriate. Future studies may extend this protocol to a larger subject group to improve statistical power and generalizability.

All participants will be briefed on the purpose of the experiment and the testing procedure before the trials begin. Participation will be voluntary, and informed consent will be obtained prior to testing.

C. Apparatus and System Configuration

The apparatus used in this study is the HapticSense belt system developed in Section III. The system consists of two HC-SR04 ultrasonic sensors, two vibration motors, an Arduino-compatible microcontroller, transistor-based motor driver circuits, base resistors, and flyback diodes. The two sensors are positioned symmetrically on the left and right sides of the wearable, forming a simplified left-right sensing structure similar to a binary spatial perception model. The motors are positioned to provide corresponding haptic feedback on the left and right sides of the waist.

This architecture is motivated by prior ultrasonic obstacle detection systems [**?**], [**?**] and vibrotactile spatial encoding studies [**?**], [**?**]. The two-channel sensing design intentionally simplifies the navigation problem to left-right directional discrimination, allowing controlled evaluation of spatial feedback effectiveness.

D. Sensor Modeling and Distance Estimation

Ultrasonic sensing in the proposed system is based on time-of-flight (ToF) measurement. The distance to an obstacle is computed as

$$d = \frac{v \cdot t}{2} \tag{1}$$

where d is the measured distance, v is the speed of sound in air, and t is the measured echo return time.

To quantify sensing performance, measured distance values will be compared with actual obstacle distances placed at known positions. The absolute distance error is defined as

$$\epsilon = |d_{measured} - d_{actual}| \tag{2}$$

To reduce measurement noise, multiple sensor readings will be averaged:

$$d_{avg} = \frac{1}{N} \sum_{i=1}^{N} d_i \tag{3}$$

where N is the number of consecutive sensor samples. This approach improves stability and reduces the influence of outliers and transient noise. Sensor limitations such as beam spread, angular reflection error, and weak reflection from small or soft obstacles will also be considered during evaluation.

E. Obstacle Detection Robustness Analysis

The system will be evaluated under different obstacle conditions in order to determine detection robustness. Obstacles of varying size, surface material, and orientation will be placed in front of the left or right sensor. For example, large flat obstacles, small objects, and angled surfaces will be tested to observe how reliably the system detects each condition.

An obstacle is considered detected when the measured distance satisfies

$$d < d_{threshold} \tag{4}$$

where $d_{threshold}$ is an experimentally selected activation threshold.

Detection reliability is quantified as

$$R = \frac{N_{detected}}{N_{total}} \tag{5}$$

where $N_{detected}$ is the number of successful detections and N_{total} is the total number of obstacle presentations.

F. Spatial Mapping and Directional Encoding

The proposed HapticSense system uses direct spatial mapping between sensing and actuation:

- Left ultrasonic sensor $\rightarrow$ Left vibration motor
- Right ultrasonic sensor $\rightarrow$ Right vibration motor

This binary mapping strategy enables intuitive interpretation of obstacle direction and is consistent with prior haptic belt research [**?**], [**?**]. The direction decision rule is defined as

$$\text{Direction} = \begin{cases} \text{Left}, & \text{if } d_L < d_{threshold},\ d_R \geq d_{threshold} \\ \text{Right}, & \text{if } d_R < d_{threshold},\ d_L \geq d_{threshold} \\ \text{Front}, & \text{if } d_L < d_{threshold},\ d_R < d_{threshold} \end{cases} \tag{6}$$

where d_L and d_R represent the measured distances from the left and right sensors, respectively.

To ensure that the mapping remains unambiguous, cross-activation error will also be monitored:

$$E_{cross} = \frac{N_{incorrect_motor}}{N_{total}} \tag{7}$$

where $N_{incorrect_motor}$ is the number of trials in which the incorrect motor is activated.

G. Vibrotactile Feedback Modeling

Two feedback modes will be examined in this study.

In the first mode, fixed-intensity vibration is applied whenever an obstacle is detected within the threshold range:

$$V = V_0 \tag{8}$$

where V_0 is a constant vibration level.

In the second mode, adaptive feedback is used so that vibration intensity increases as the obstacle becomes closer. This can be modeled as

$$V(d) = k \cdot \frac{1}{d} \tag{9}$$

where $V(d)$ is the vibration intensity and k is a scaling constant.

For implementation on the embedded platform, this relationship may be discretized using duty-cycle bands, for example:

- $d > 80$ cm: motor OFF

- $50 < d \leq 80$ cm: weak vibration
- $30 < d \leq 50$ cm: moderate vibration
- $d \leq 30$ cm: strong vibration

This modulation strategy is intended to encode urgency and improve user response speed, which directly supports the investigation of Hypothesis H2.

H. System Latency Analysis

The total latency of the system is modeled as

$$T_{total} = T_{sensor} + T_{processing} + T_{actuation} \tag{10}$$

where T_{sensor} is the ultrasonic measurement delay, $T_{processing}$ is the microcontroller computation time, and $T_{actuation}$ is the motor activation delay.

Latency will be estimated using timestamps at the sensor trigger event and the corresponding motor activation event:

$$T_{latency} = t_{motor} - t_{trigger} \tag{11}$$

Low latency is important because the system is intended for real-time obstacle awareness.

I. Experimental Procedure

The study will be conducted in a controlled indoor environment. Each participant will wear the HapticSense belt at waist level. The sensors and motors will be calibrated before each test session to ensure consistent placement and operation.

For each trial, an obstacle will be introduced at a predefined position corresponding to the left sensor, right sensor, or both sensors simultaneously. The participant will either stand or move slowly under controlled conditions while the device detects the obstacle and generates haptic feedback.

Each trial will follow the sequence below:

1) Initialize the system and calibrate sensors
2) Place obstacle at a predefined direction and distance
3) Acquire ultrasonic sensor readings
4) Determine obstacle direction using threshold logic
5) Activate the appropriate vibration motor pattern
6) Record the participant response and system timing
7) Repeat across multiple trials and conditions

The flow of the experimental protocol is illustrated in Fig. 2.

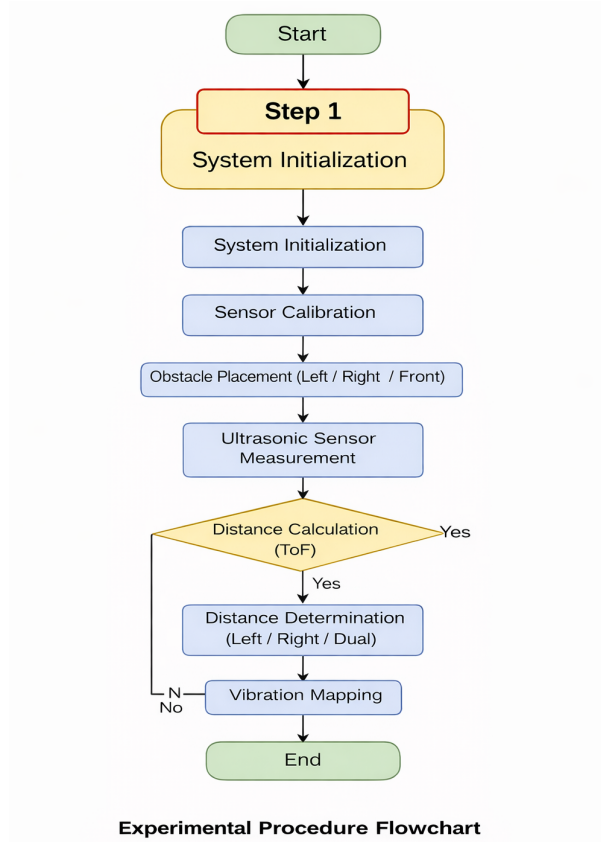


Fig. 2. Flow diagram of the experimental procedure used to evaluate the HapticSense system, from initialization and calibration through obstacle presentation, motor activation, user response, and repeated trials.

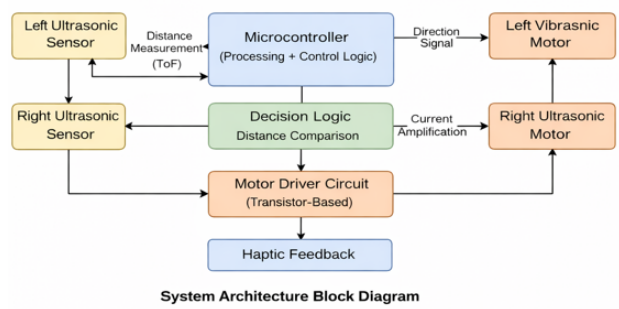


Fig. 3. Block diagram of the HapticSense methods framework showing left and right ultrasonic sensing, microcontroller-based distance processing, decision logic, transistor motor driving, and directional haptic output.

J. Block Diagram of the Methods Framework

The system-level methods framework is shown in Fig. 3. The block diagram summarizes the interaction between sensing, processing, decision logic, motor driving, and haptic output. It also clarifies how left and right sensor data are fused through the control algorithm before vibrotactile actuation.

K. Algorithmic Procedure

The control logic used in the methods plan is summarized in Algorithm IV-K.

[t] HapticSense Obstacle Detection and Feedback Logic [1] Initialize system Set threshold distance $d_{threshold}$ system is active Read left sensor distance d_L Read right sensor distance d_R Filter readings using moving average $d_L < d_{threshold}$ and $d_R \geq d_{threshold}$ direction $\leftarrow$ LEFT $d_R < d_{threshold}$ and $d_L \geq d_{threshold}$ direction $\leftarrow$ RIGHT $d_L < d_{threshold}$ and $d_R < d_{threshold}$ direction $\leftarrow$ FRONT direction $\leftarrow$ NONE direction = LEFT Activate left motor Deactivate right motor direction = RIGHT Activate right motor Deactivate left motor direction = FRONT Activate both motors Deactivate both motors adaptive mode enabled Compute intensity using nearest detected distance Apply PWM modulation Record distance, direction, and response timing

L. Human Response Measurement

Three primary response measures will be collected.

First, reaction time is defined as

$$RT = t_{response} - t_{stimulus} \tag{12}$$

where $t_{stimulus}$ is the onset of vibration and $t_{response}$ is the participant response time.

Second, directional accuracy is computed as

$$Accuracy = \frac{N_{correct}}{N_{total}} \times 100 \tag{13}$$

Third, perceptual consistency for dual-motor activation is computed as

$$Consistency = \frac{N_{correct_interpretation}}{N_{dual_trials}} \tag{14}$$

These measures directly correspond to Hypotheses H1, H2, and H3, respectively.

M. Statistical Analysis Plan

Because this stage of the project focuses on proof-of-concept and small-sample evaluation, the statistical analysis will primarily use descriptive measures. Mean response time is computed as

$$\mu = \frac{1}{N} \sum RT_i \tag{15}$$

and standard deviation is computed as

$$\sigma = \sqrt{\frac{1}{N} \sum (RT_i - \mu)^2} \tag{16}$$

To compare fixed and adaptive vibration conditions, the change in response time is defined as

$$\Delta RT = RT_{fixed} - RT_{adaptive} \tag{17}$$

The hypothesis evaluation criteria are:

- H1: directional accuracy $> 50\%$
- H2: $RT_{adaptive} < RT_{fixed}$
- H3: consistency $> 70\%$

Although these analyses are relatively simple, they provide an appropriate quantitative basis for evaluating the hypotheses in a prototype-stage study.

N. Calibration and Pilot Testing

Before the main trials begin, pilot testing will be conducted to verify sensor calibration, motor activation, threshold selection, and overall system reliability. Calibration will be performed at known obstacle distances to confirm that ultrasonic readings remain stable and that threshold-based activation behaves consistently.

Pilot testing also helps identify issues such as false activation, inconsistent reflection from small obstacles, or unstable vibration output. These preliminary trials ensure that the full experimental procedure is feasible and repeatable.

O. Replicability and Generalization

The methods described here are intended to be systematic, transparent, and reproducible. All key variables, processing logic, calibration steps, and evaluation metrics are explicitly defined so that the study can be replicated by another researcher under similar conditions. Although the present system uses only two sensors and two actuators, the methodology is modular and can be extended to future multi-sensor wearable configurations.

References

[1] F. A. Sharom, S. A. Che Kar, R. Abdullah, S. Mohamed, and N. Ahmed, "Design and Implementation of an Arduino-Based Wearable Assistive System for Obstacle Detection in Blind Navigation," in *Proc. 2025 IEEE International Conference on Applied Electronics and Engineering (ICAEE)*, Shah Alam, Malaysia, 2025, pp. 1–6, doi: 10.1109/ICAEE65866.2025.11170895.

[2] A. Cosgun, E. A. Sisbot, and H. I. Christensen, "Evaluation of rotational and directional vibration patterns on a tactile belt for guiding visually impaired people," in *Proc. 2014 IEEE Haptics Symposium (HAPTICS)*, Houston, TX, USA, 2014, pp. 367–370, doi: 10.1109/HAPTICS.2014.6775483.

[3] A. T. Olomowewe, E. O. Joseph, S. U. Hussein, S. Thomas, M. C. Chukwu, and P. Akah, "Design and Implementation of a Wearable Device for Obstacle Detection and Warning," in *Proc. 2021 1st International Conference on Multidisciplinary Engineering and Applied Science (ICMEAS)*, Abuja, Nigeria, 2021, pp. 1–4, doi: 10.1109/ICMEAS52683.2021.9692365.

[4] G. Flores *et al.*, "Vibrotactile Guidance for Wayfinding of Blind Walkers," *IEEE Transactions on Haptics*, 2015.